

Xinhao Tan

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EDUCATION

Carnegie Mellon University

Pittsburgh, PA, USA

M.S. in Information Networking; GPA: 3.67/4.00

Aug 2025 – Dec 2026

- **Relevant coursework:** Database Systems, Distributed Systems, Computer Networks, AI Infrastructure

National University of Singapore

Singapore

B.Eng. in Electrical Engineering; GPA: 4.73/5.00

Aug 2021 – Aug 2025

- Minor in Artificial Intelligence; Track: Internet of Things (IoT)

SKILLS

Programming Languages: Python, Java, C, C#, Go, MATLAB, SQL, JavaScript, CSS, Bash

AI/ML Toolchains: PyTorch, Faiss, DeepSpeed, HuggingFace, PEFT/LoRA, LangChain, RAG, SGLang, vLLM

Systems & DevOps: Docker, Kubernetes, Nginx, Git/GitHub Actions, Linux, Redis, MySQL, Kafka, RabbitMQ, Spring Boot, Flask, AWS

EXPERIENCE

Aruvii

Singapore

3D Object-Tracking Researcher (Seasonal)

Aug 2024 – May 2025

- Researched a LiDAR-based 3D multi-object tracking algorithm for human security applications; investigated both filter-based and learning-based methods to improve robustness in crowded environments.
- Engineered a real-time tracking system using **ROS** and **RViz**, integrating Kalman-filter prediction and Hungarian-algorithm data association within a Fast-Poly framework.
- Implemented an Interacting Multiple Model (**IMM**) filter with Constant Velocity (**CV**) and Constant Turn Rate (**CVTR**) motion models, achieving an **11%** improvement in tracking accuracy for non-linear object motion.

PUBLICATIONS

Open-world Authorship Attribution – ACL 2025

First author

- Designed an **LLM-based pipeline** for multi-hierarchy keyword generation to characterize anonymous context. Implemented candidate paper search and collection via scholar search engines with web scraping. Built an author decision-making framework using LLMs with signals such as writing style, dataset usage, and contributions; curated **300+** anonymous papers for evaluation. achieving **60.7%** and **44.3%** accuracy in the two stages, respectively

PROJECTS

Event Extraction Using Large Language Models

May 2025 – Aug 2025

- Built an end-to-end event type classification pipeline (label mapping, inference, evaluation) for short financial messages; handled ambiguous labels with self-consistency and **multi-vote** decoding over prompt variants.
- Added **RAG**-based dynamic few-shot prompting (**LangChain** + similarity retrieval of labeled exemplars); achieved **0.60+** Macro-F1 and improved tail-class stability vs. prompt-only baseline.
- Finetuned **Qwen-3** 4B with **LoRA** and scaled training with **DeepSpeed** multi-GPU (**ZeRO-2/3**); achieved **0.75+** Macro-F1, yielding **+300%** absolute points over the base model.

Payment E-commerce System

May 2025 – Aug 2025

- Developed a small-scale payment e-commerce system. Built backend services with **Spring Boot/Spring/MyBatis** integrated WeChat OAuth and Alipay sandbox for order creation, payment handling, and async callbacks.
- Integ highrated **RabbitMQ** (exchanges/queues, routing keys, DLQ) for buffering high-volume callbacks; used per-message ack and delayed retries for idempotent updates.
- Deployed with **Docker**, fronted by **Nginx**; exposed **15+** REST APIs and designed schema for **10k+** orders.

Machine Unlearning with TOFU Dataset

Feb 2024 – May 2025

- Built an end-to-end unlearning pipeline on TOFU with a fine-tuned **Phi-1.5**: GradDiff-style negative gradient ascent (**AdamW**). added curvature-aware updates via **Sophia** (diagonal Hessian + clipping) for stable forgetting.
- Designed localized unlearning: selective updates on attention projections, **layer sensitivity** ranking via gradient norms (Top-*K*) with targeted freezing, plus **token-level** influence and selective masking to preserve utility.
- Scaled experiments with **DeepSpeed ZeRO** + **bf16**, deterministic seeding, and experiment tracking; achieved **95.8%** Forgetting Quality (+17 pts vs. baseline) with quantified Model Utility trade-offs.